
MARKET MYSTERIES, PART 1

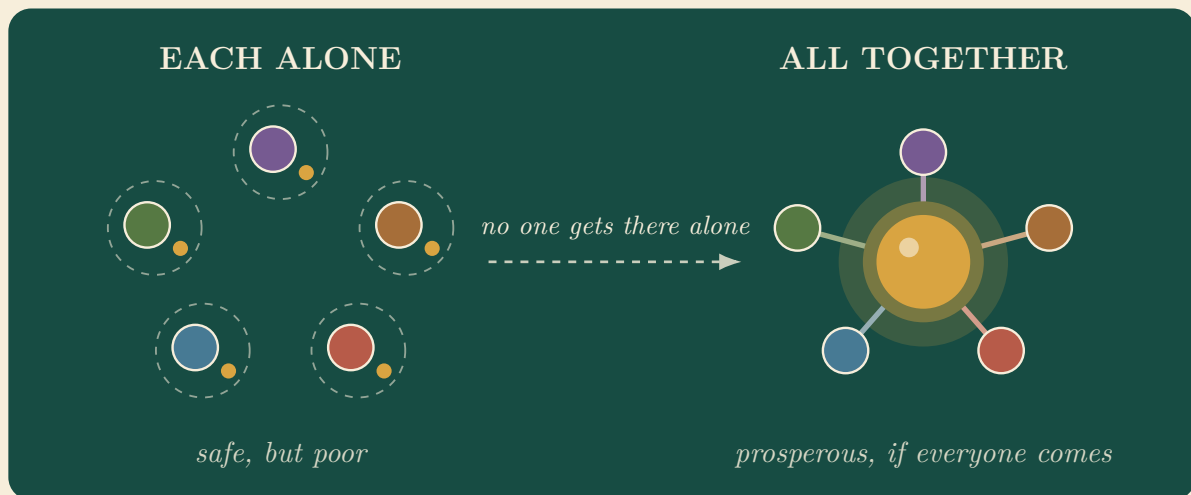
Does individual rationality guarantee collective optimality?

Prosper Together

A Tale of Specialization, Coordination and Cooperation

The model behind the game

Production, equilibria, survival,
learning automata and pledges



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with the help of AI/SI friends

Companion paper

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Abstract

Prosper Together puts one human player among automata who each divide three units of energy per turn between safe production at home and a shared table at which their different trades are combined. Energy kept at home yields one unit of wellbeing. Energy sent to the table yields nothing if the player is alone there, and up to three and a half times as much if everybody takes part in equal measure. The one-turn game has exactly two pure-strategy Nash equilibria, universal autarky and universal full commitment, and the second is better for everyone. This paper states the rules of the game precisely and derives their consequences. It covers the production technology and its equilibria for five and for seven players; the stock of durable goods, consumption and the survival rule that turn the one-turn game into a dynamic one; the automata, which forecast each other player's contribution separately, learn at different constant rates and choose by expected value subject to survival; and the pledge, a payment from the human player's reserve that buys an automaton's additional energy for one turn at a price the automaton sets. The society starts in autarky with everyone expecting autarky. The player's task is to shift expectations far enough that cooperation sustains itself without payment. A final section analyzes the institutional alternative the game proposes: a democratically elected centralized coordinator that guarantees a minimum return at the table.

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1 Introduction

1.1 Motivation

Free markets are often considered the best available way to allocate and use resources. They are also often seen as the natural state of things, or simply as the absence of regulation and state involvement beyond the enforcement of individual property rights. The *Market Mysteries* series sets out to illustrate that free markets cannot be universally recommended, because what decentralized decisions lead to depends entirely on the underlying economic environment.

This first part, the game *Prosper Together*, captures one core aspect of economic reality. Individuals possess unique qualities that make specialization beneficial to the whole. But specialization is risky: whoever specializes depends on the whole to provide necessities. A baker who bakes only bread needs a farmer, a miller and customers; if they do not show up, the baker is worse off than someone who kept a garden. The game makes this dependence literal. A player who practices a trade alone produces nothing of value, and a player who stays home is safe but poor.

1.2 Related setups

Prosper Together is an adaptation of the *stag hunt*. Rousseau [25] describes hunters who can bring down a deer only together, while each of them can catch a hare alone and thereby abandons the others. Hunting the stag is best for all if all take part, but hunting hare is the safe choice, and both “everyone hunts stag” and “everyone hunts hare” are self-confirming. Skyrms [28] argues that this, rather than the prisoner’s dilemma, is the right model of the social contract: the problem is not that cooperation is against anyone’s interest, but that it requires trust in the others’ cooperation.

The same structure appears throughout economics under other names.

- **Coordination failures.** With strategic complementarities, where one agent’s effort raises the return to another’s, an economy can have several equilibria ranked by welfare, and can be stuck in a bad one although nobody makes a mistake. Bryant [3] shows this for team production in which output is limited by the smallest input, Diamond [9] for trade that requires finding a partner, and Cooper and John [7] give the general argument; Cooper [6] surveys the field. In the laboratory, large groups playing Bryant’s game reliably slide to the worst equilibrium [31].
- **The big push.** Rosenstein-Rodan [24] observed that a single modern factory may be unprofitable in a poor economy because its workers cannot buy all of its output, while many factories in different industries, started together, would be each other’s customers. Murphy, Shleifer and Vishny [22] formalized the idea: industrialization and its absence are both equilibria, and moving from one to the other takes a coordinated investment.
- **Complementary tasks.** In Kremer’s O-ring theory [17] the value of a product is destroyed if any one of its tasks is done badly, so the return to one worker’s skill depends on the skill of all co-workers.
- **The division of labour.** That specialization raises productivity, and that it is limited by the extent of the market, goes back to Smith [29]; Young [33] turned it into a theory of increasing returns, and Becker and Murphy [1] argue that the binding limit is the cost of coordinating specialists.
- **Which equilibrium?** Harsanyi and Selten [13] distinguish the payoff-dominant equilibrium, which is better for everyone, from the risk-dominant one, which is the better reply to an uncertain opponent. Models of adaptive play with occasional mistakes [16, 32] and of small payoff uncertainty [4] select the risk-dominant equilibrium, and Krugman [18] and Matsuyama [21] ask when history and when expectations decide.

- **Learning.** How players come to play an equilibrium is the subject of the literature on learning in games [12]. In fictitious play [2] each player best-responds to the observed frequencies of the others' past actions; weighting recent observations more heavily fits experimental behaviour better [5], and such adaptive dynamics explain the laboratory slide into bad equilibria [8].
- **Getting out.** A leader can change what followers expect by acting first at his own cost [14]. An entrepreneur can make contributing a dominant strategy by promising to compensate contributors if too few others contribute [30]. And an institution can remove a bad equilibrium by insuring against it, at no cost if the insurance is believed, as deposit insurance does for bank runs [10].

1.3 What the game adds, and the plan of the paper

The game takes the stag hunt and adds what a society that actually tries to leave autarky has to deal with. There are more than two players, and they are not interchangeable: the return at the table depends on how many different trades take part and on how evenly. Commitment comes in degrees. Players have to eat: a turn that goes badly is not just a low payoff but a step towards starvation, and a player who starves takes a trade out of the economy for good. The other players are rational: each does what is best for itself given what it expects of the others. But nobody can know what the others will do. Expectations have to be formed from what has been seen, and the players revise them at different speeds. And there is an instrument: the human player can pay others to take part, out of savings, and thereby change what everyone else observes.

Section 2 defines production and payoffs. Section 3 derives the equilibria of the one-turn game and asks how much assurance a player needs to commit. Section 4 adds goods, consumption and survival. Section 5 describes the automata, and Section 6 the pledges. Section 7 states the player's objective and the forces that work for and against it. Section 8 analyses the institutional alternative, and Section 9 states what the model does not claim. All parameter values are collected in Appendix A.

2 The stage game

2.1 Players and trades

There are n players, each with a trade of their own. The game is played with $n = 5$ (Level 1) and with $n = 7$ (Level 2). One player is human; in the game this is the Artist, and this paper calls that player the *instigator*. The others are automata.

Player	Trade	Present in
Sable	Mystic	both levels
Marlow	Farmer	both levels
Iona	Scientist	both levels
the instigator	Artist	both levels
Ansel	Craftsperson	both levels
Mira	Healer	Level 2
Rowan	Organizer	Level 2

The trades enter the payoff function symmetrically. Their role is to make each player's contribution *different in kind* from every other player's, so that what counts at the table is the number of distinct contributors and the balance between them, not the total amount contributed. The automata do differ in how fast they learn (Section 5).

2.2 Energy and the two technologies

At the start of every turn each living player receives three indivisible units of energy. Player i chooses how many to commit to the table,

$$c_i \in \{0, 1, 2, 3\},$$

and uses the remaining $3 - c_i$ at home.

Home production (autarky). Every unit of energy used at home produces goods worth exactly one unit of wellbeing, whatever anyone else does.

The table (specialization). Every unit of energy committed to the table becomes an input of its owner's trade. An input is worth nothing by itself. If only one player has committed energy, it is wasted. If at least two players have, their inputs are combined into goods, and every unit of energy at the table earns its owner the same return, which is higher the more trades take part and the more evenly they do so.

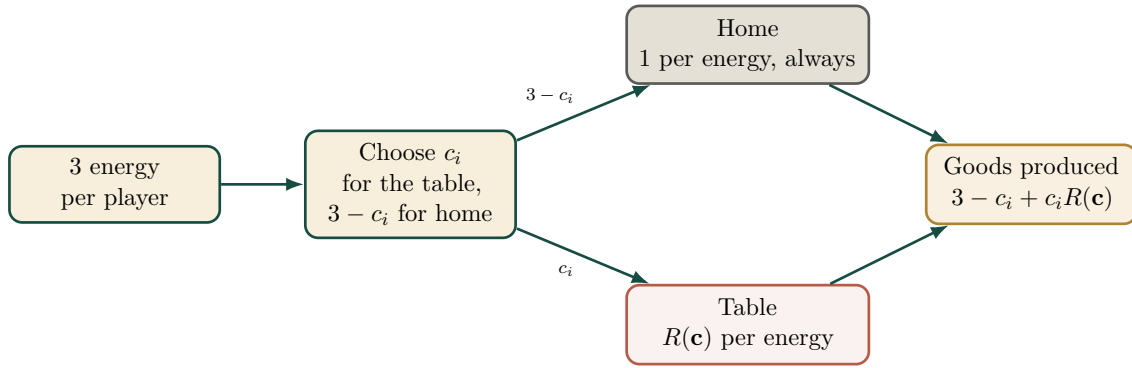


Figure 1: Production in one turn. The home return is certain. The table return depends on the whole profile of commitments.

2.3 Payoffs

Let $N = \{1, \dots, n\}$ and let $\mathbf{c} = (c_1, \dots, c_n) \in \{0, 1, 2, 3\}^n$ be a profile of commitments. Define

$$\begin{aligned}
 T(\mathbf{c}) &= \sum_{i \in N} c_i && \text{(total energy at the table),} \\
 Q(\mathbf{c}) &= \sum_{i \in N} c_i^2 && \text{(its concentration),} \\
 m(\mathbf{c}) &= \#\{i : c_i > 0\} && \text{(number of contributing trades).}
 \end{aligned}$$

When at least one player contributes, the *effective number of trades* is

$$H(\mathbf{c}) = \frac{T(\mathbf{c})^2}{Q(\mathbf{c})}. \quad (1)$$

This is the inverse of the Herfindahl concentration index of the energy at the table. By the Cauchy–Schwarz inequality $1 \leq H(\mathbf{c}) \leq m(\mathbf{c}) \leq n$. If k players commit equal amounts, $H(\mathbf{c}) = k$; unequal commitments lower it. The return per unit of energy at the table is

$$R(\mathbf{c}) = \begin{cases} 0, & m(\mathbf{c}) < 2, \\ 3.5 \left(\frac{H(\mathbf{c})}{n} \right)^{1.5}, & m(\mathbf{c}) \geq 2, \end{cases} \quad (2)$$

and player i produces goods worth

$$u_i(\mathbf{c}) = \underbrace{(3 - c_i)}_{\text{home}} + \underbrace{c_i R(\mathbf{c})}_{\text{table}}. \quad (3)$$

Three properties follow directly.

1. *Nobody can specialize alone.* A lone contributor gets nothing for the energy committed. This is the complementarity at the heart of the game.
2. *The table is shared in proportion.* All contributors earn the same rate, so each one's share of the table's output is proportional to the energy he or she put in. A unit at the table beats a unit at home exactly when $R(\mathbf{c}) > 1$.
3. *Breadth and balance both matter.* The rate rises more than proportionally with the effective number of trades (the exponent is 1.5), and it reaches its maximum of 3.5 only when all n trades contribute equally. The denominator is the number of trades in the society, not the number at the table: a trade that stays away is missed.

2.4 Critical mass

If k players commit equally and the others stay home, the rate is

$$R_k = 3.5 \left(\frac{k}{n} \right)^{1.5} \quad (k \geq 2), \quad (4)$$

which exceeds the home return of 1 only if $k > 3.5^{-2/3} n \approx 0.434 n$.

Equal contributors k	1	2	3	4	5	6	7
R_k with $n = 5$	0	0.885	1.627	2.504	3.500		
R_k with $n = 7$	0	0.535	0.982	1.512	2.113	2.777	3.500

Table 1: Return per unit of energy at the table when k trades contribute equally. Bold: the smallest group that beats autarky.

With five players a group of three is the smallest that does better than staying home; with seven it takes four. Two players alone always lose by cooperating. The larger society is therefore harder to move: more trades are essential, and a coalition of a given size is worth less.

2.5 Worked profiles

The profiles below are for $n = 5$, in the order (Sable, Marlow, Iona, instigator, Ansel). $W(\mathbf{c}) = \sum_i u_i(\mathbf{c})$ is total production.

Situation	$\mathbf{c}$	H	R	Individual production	W
Universal autarky	(0, 0, 0, 0, 0)	–	0	3 each	15
A lone contributor	(0, 0, 0, 1, 0)	1	0	2 for the contributor	14
Two, unbalanced	(1, 0, 0, 3, 0)	1.6	0.634	2.63 and 1.90	13.53
Three, unbalanced	(1, 1, 0, 3, 0)	2.273	1.073	3.07, 3.07, 3.22	15.36
Three, balanced	(3, 0, 3, 3, 0)	3	1.627	4.88 each	20.64
Universal full commitment	(3, 3, 3, 3, 3)	5	3.5	10.5 each	52.5

Table 2: Representative outcomes in the five-player game. Players who stay home produce 3.

The third row is what a first attempt typically looks like: the instigator commits everything, one other player contributes a single unit, and both earn less at the table than they would have at home.

3 Equilibria of the stage game

This section considers one turn in isolation, with all players alive, no payments between players, and each player maximizing his or her own production u_i . A profile $\mathbf{c}^*$ is a pure-strategy Nash equilibrium if nobody gains by changing only his or her own commitment:

$$u_i(c_i^*, \mathbf{c}_{-i}^*) \geq u_i(d, \mathbf{c}_{-i}^*) \quad \text{for all } i \in N \text{ and } d \in \{0, 1, 2, 3\}.$$

3.1 Universal autarky

Let $\mathbf{c}^A = (0, \dots, 0)$. Everybody produces 3. A player who deviates alone to $d \geq 1$ is the only contributor, gets nothing at the table, and produces $3 - d < 3$. Staying home is the unique best reply to everybody else staying home, so universal autarky is a *strict* Nash equilibrium. It is self-enforcing because the first step out of it is costly and has to be taken alone.

3.2 Universal full commitment

Let $\mathbf{c}^F = (3, \dots, 3)$. Commitments are perfectly balanced, so $H = n$, $R = 3.5$ and everybody produces 10.5. If the other $n - 1$ players commit 3 and player i commits d , then

$$H_i(d) = \frac{(3(n-1) + d)^2}{9(n-1) + d^2}.$$

	$n = 5$			$n = 7$		
d	$H_i(d)$	R	u_i	$H_i(d)$	R	u_i
0	4.000	2.504	3.000	6.000	2.777	3.000
1	4.568	3.056	5.056	6.564	3.178	5.178
2	4.900	3.396	7.791	6.897	3.423	7.845
3	5.000	3.500	10.500	7.000	3.500	10.500

Table 3: Unilateral deviations from universal full commitment. For $d = 0$ the rate is that of the remaining table; the deviator has no part in it.

Committing everything is the unique best reply to everybody else committing everything, so universal full commitment is a strict Nash equilibrium too. Note that a deviator hurts the others as well as himself: withdrawing lowers the rate for everyone who stays.

3.3 There are no other pure equilibria

The game is finite, so this can be checked exhaustively. For every profile and every player, all three alternative commitments are compared with the one in the profile:

	Profiles	Deviations checked	Pure Nash equilibria
$n = 5$	$4^5 = 1,024$	15,360	$(0, \dots, 0)$ and $(3, \dots, 3)$
$n = 7$	$4^7 = 16,384$	344,064	$(0, \dots, 0)$ and $(3, \dots, 3)$

Intermediate symmetric profiles are instructive. Everybody committing one unit achieves the full effective number of trades and a production of 5.5 each, but it is not an equilibrium: a player who raises his commitment to two produces about 6.98 ($n = 5$) or 7.12 ($n = 7$). Against everybody committing two, raising to three lifts production from 8 to about 10. Once the table pays, the incentive is to put in more, and the only resting point is full commitment.

3.4 Welfare

Both equilibria are strict, but they are not equivalent. Every player produces 3 in autarky and 10.5 at full commitment, three and a half times as much. Full commitment strictly Pareto-dominates autarky. The game therefore separates two questions that are easily confused: whether cooperation is individually rational once it is established (it is, with no enforcement needed), and whether individually rational behaviour gets a society there (not from autarky).

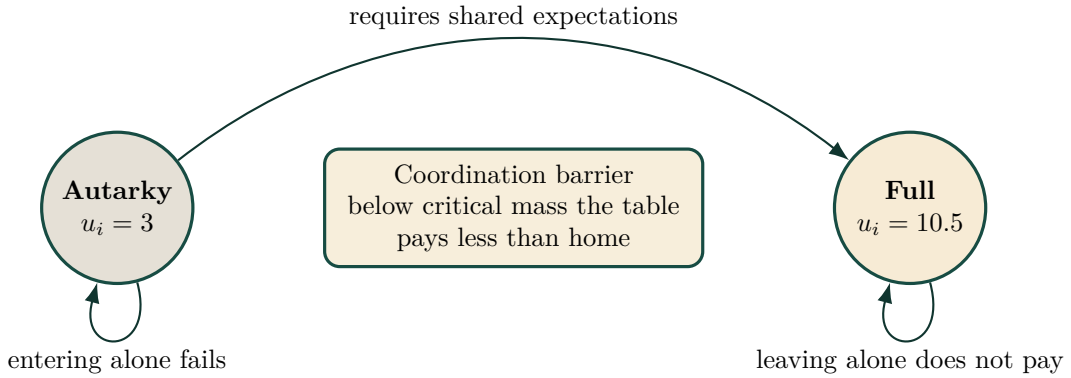


Figure 2: Two self-confirming states. The gap between them is a problem of coordination, not of incentives at the cooperative state.

3.5 How much assurance does a player need?

Two calculations describe the barrier more precisely.

Counting committed partners. Suppose k of the other players commit everything and the rest stay home. The best reply is to stay home if k is too small, and to commit everything as soon as k is large enough; partial commitment is never best.

Others fully committed, k	0	1	2	3	4	5	6
$n = 5$: production from $c_i = 3$	0	2.66	4.88	7.51	10.50		
$n = 7$: production from $c_i = 3$	0	1.60	2.95	4.54	6.34	8.33	10.50

Table 4: Production from committing everything when k others do; staying home always yields 3. Bold: the smallest k at which committing is the best reply.

The tipping point is two committed partners with five players and three with seven: the critical mass of Table 1, counting oneself.

Uncertain partners. Suppose instead that a player believes each of the others, independently, commits everything with probability q and stays home otherwise. The expected production from committing everything is

$$\mathbb{E} u_i(3) = 3 \sum_{k=0}^{n-1} \binom{n-1}{k} q^k (1-q)^{n-1-k} R_{k+1}, \quad (5)$$

with $R_1 = 0$ and R_k from Equation (4). It exceeds the 3 from staying home when $q > q^*$, where

$$q^* \approx 0.296 \ (n = 5), \quad q^* \approx 0.327 \ (n = 7).$$

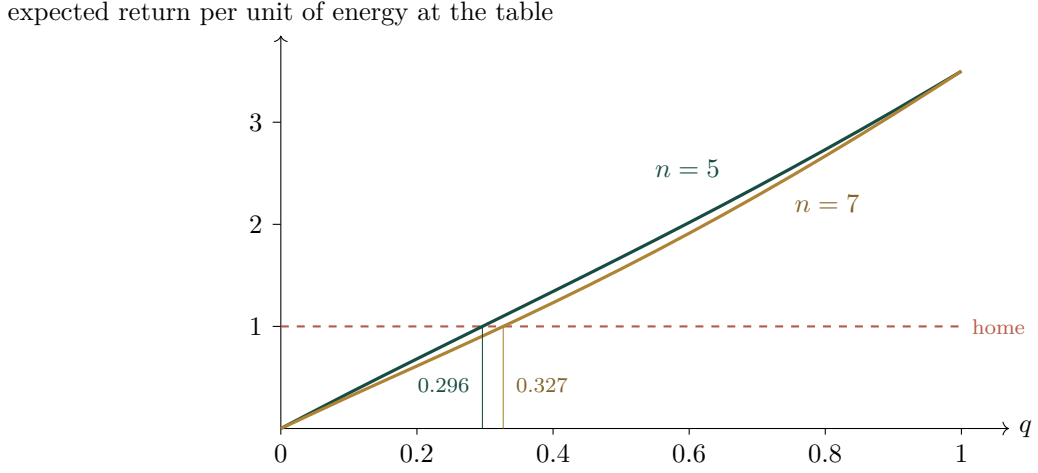


Figure 3: Expected table return when each other player commits everything with probability q . Committing pays in expectation to the right of the break-even belief.

The break-even belief is well below one half. A player who thought each of the others as likely to commit as not would expect about 5.0 ($n = 5$) or 4.7 ($n = 7$) from committing, against 3 from staying home. In this sense full commitment is not only the better equilibrium but also the less demanding one in terms of beliefs. The trap is therefore not that autarky is the prudent choice under uncertainty. It is that a history of autarky does not leave beliefs uncertain: everybody has seen everybody stay home, and expects that. The game starts in exactly this state.

4 Goods, consumption and survival

The stage game is played turn after turn, and three things connect the turns: goods can be stored, players must consume, and players who consume too little die.

4.1 Stocks and flows

Player i enters turn t with a reserve $b_{it} \geq 0$ of durable goods. Let $G_{it} = u_i(\mathbf{c}_t)$ be the goods produced in turn t and P_{it} the net payment received before production (Section 6; negative for the instigator, positive for a paid automaton, zero otherwise, and $\sum_i P_{it} = 0$). The goods available after production are

$$A_{it} = b_{it} + P_{it} + G_{it}. \quad (6)$$

The player then chooses how much to consume, $0 \leq z_{it} \leq A_{it}$, and keeps the rest:

$$b_{i,t+1} = A_{it} - z_{it}. \quad (7)$$

One good consumed is one unit of wellbeing. Goods in reserve do not spoil and earn nothing. There is no borrowing: a reserve cannot be negative, and payments can only be made out of goods already held. Every player starts with the same reserve $b_{i1} = b_0$.

4.2 The survival rule

Wellbeing has to be kept up. With a window of L turns and a minimum M , player i survives turn t only if

$$\sum_{r=t-L+1}^t z_{ir} \geq M \quad \text{for every } t \geq L. \quad (8)$$

Only full windows are judged, so the first verdict falls at the end of turn L , and equality is enough. The rule applies to the instigator and to every automaton alike. In Level 1, $M = 6$ and $L = 2$; in Level 2, $M = 9$ and $L = 3$. Both ask for an average of three per turn, which is exactly what autarky produces.

What counts is consumption in the window, not wealth and not the past. Goods in reserve protect a player only when they are eaten, and a good turn drops out of the window after L turns.

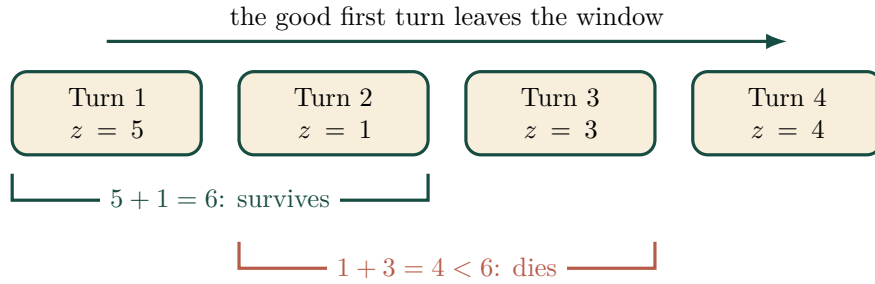


Figure 4: The rolling window with $M = 6$, $L = 2$. A total of nine over three turns does not help: the second window is short.

The consumption a player needs in turn t to pass that turn's check is

$$d_{it} = \begin{cases} 0, & t < L, \\ \max\{0, M - \sum_{r=t-L+1}^{t-1} z_{ir}\}, & t \geq L. \end{cases} \quad (9)$$

If $A_{it} < d_{it}$ the player cannot be saved. Otherwise consumption is restricted to $d_{it} \leq z_{it} \leq A_{it}$: nobody may save goods that are needed to survive the current turn.

4.3 Autarky feeds a player but does not make one rich

A player in autarky produces three goods per turn and must consume an average of three. Over any L consecutive turns production equals the required consumption, so the reserve at the end of a window can never be larger than at its start. Autarky is sustainable indefinitely, but it yields no surplus. The starting reserve is the only wealth an autarkic society has, and once it is spent it is not replaced. Whatever is to be paid for a second time has to be paid for out of what the table has yielded.

A player who falls behind is in a different position. After consuming only two in a turn of Level 1, the next turn needs four, which home production alone cannot provide. The player needs a reserve, or a table that pays.

4.4 Death and the shrinking frontier

An automaton that fails the survival check is removed for the rest of the run. It receives no more energy, produces and consumes nothing, and its trade is lost. If the instigator fails, the run is over.

Production after a death uses Equations (1)–(2) with the dead players’ commitments set to zero. The denominator n in the rate does *not* shrink: the society still lacks the trade. With s survivors the best possible rate is

$$R_{\max}(s) = 3.5 \left(\frac{s}{n} \right)^{1.5} \quad (s \geq 2), \quad (10)$$

the values of Table 1. One death lowers the best achievable production per player from 10.5 to 7.5 ($n = 5$) or 8.3 ($n = 7$). With two survivors of five, or three or fewer of seven, even perfect cooperation pays less than staying home, and the prosperous equilibrium no longer exists.

4.5 Survival and attitudes to risk

Nothing in the rules makes a player risk-averse; goods are valued one for one. But the survival threshold makes the consequences of a gamble depend on the player’s state. A player whose requirement can still be met from home production has something to lose at the table. A player who needs more than a reserve plus three goods cannot be saved by staying home, and for that player the table is the only chance. Threshold models of this kind are known to produce caution in some states and risk-seeking in others [26, 19]; which one obtains depends on the distance to the threshold and on the outcomes available.

4.6 The score

A run is scored by average wellbeing per turn, counting what is left in reserve at the end:

$$s_i = \frac{\sum_{t=1}^{\tau} z_{it} + b_{i,\tau+1}}{\tau} \quad (11)$$

after τ completed turns. Dividing by the number of turns makes runs of different length comparable, and counting the final reserve means that saving is not penalized. The score plays no role in the survival rule.

5 The automata

The automata are not scripted. Each one keeps a forecast of what every other player will contribute, updates it from what it observes, and chooses the commitment that is best for itself given that forecast and its own survival. They differ only in how fast they update. Section 5.6 discusses in what sense this makes them rational.

5.1 Forecasts

Automaton j holds, for every other player i (including the instigator), a separate probability distribution $\pi_{ji,t}$ over i ’s commitment in turn t , that is, over $\{0, 1, 2, 3\}$. All forecasts start from the same distribution $\pi_{j \cdot, 1} = \pi_0$, and in both levels

$$\pi_0 = (1, 0, 0, 0) :$$

every automaton is certain that nobody will contribute anything. After each turn, having seen that i committed c_{it} , automaton j moves its forecast towards what it saw at its own learning rate $\eta_j \in [0, 1]$:

$$\pi_{ji,t+1}(k) = (1 - \eta_j) \pi_{ji,t}(k) + \eta_j \mathbf{1}\{c_{it} = k\}, \quad k = 0, 1, 2, 3. \quad (12)$$

Several things follow.

- *Zero is evidence.* A player who stays home is observed staying home, and the forecast moves towards zero. Trust that is not confirmed decays.
- *Memory fades geometrically.* The latest observation has weight η_j , and an observation s turns old has weight proportional to $(1 - \eta_j)^s$. The automaton never becomes set in its ways: after any history, enough new evidence overturns the forecast. With $\eta_j = 0$ beliefs never change; with $\eta_j = 1$ only the last turn counts.
- *Certainty of autarky is not a trap.* Although the initial forecast puts probability zero on any contribution, one observed contribution of k gives category k probability η_j .
- *Only commitments are observed.* The forecast uses what each player put on the table and nothing else: not what the table yielded, not what anybody consumed, and not whether a contribution was voluntary or paid for (Section 6).
- *Each peer is judged separately.* That one contributor disappears says nothing about the others.
- *The dead contribute nothing.* A dead player is forecast at zero with certainty.

Rule (12) is fictitious play with exponentially declining weights on the past [2, 5], or equivalently adaptive expectations applied to each category of each peer's behaviour. It is not Bayesian updating of a stationary frequency, whose step size would shrink as observations accumulate.

The probability that i contributes at all, as j sees it, is $q_{ji,t} = 1 - \pi_{ji,t}(0)$. If i contributes for s turns in a row starting from π_0 , then $q = 1 - (1 - \eta_j)^s$; if i then stops, q shrinks by the factor $(1 - \eta_j)$ per turn.

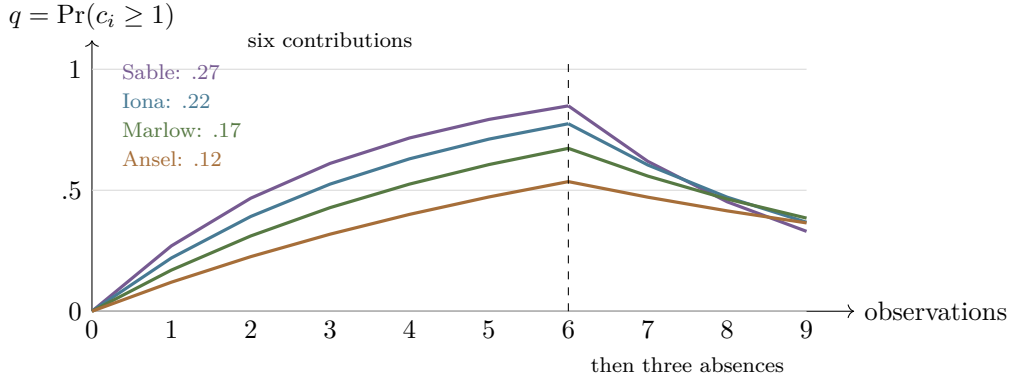


Figure 5: What four automata with different learning rates believe about one peer who contributes six turns in a row and then stops. Fast learners are won quickly and lost quickly.

5.2 Choice

Automaton j treats the other players' commitments as independent, so its forecast of the whole profile is

$$\Pr_{jt}(\mathbf{c}_{-j}) = \prod_{i \neq j} \pi_{ji,t}(c_i). \quad (13)$$

For each of its own four options it computes the production it would obtain in *every* combination of the others' commitments (4^{n-1} combinations: 256 with five players, 4,096 with seven) and weights each by its probability. The enumeration is exact. Evaluating the production function at the others' *average* commitments instead would be wrong, because the rate is nonlinear and the automaton's own commitment changes both the breadth and the balance of the table.

Outcomes are valued by the goods they leave the automaton with, except that an outcome in which

it cannot survive is worth nothing:

$$v_{jt}(x) = \begin{cases} x, & x \geq d_{jt}, \\ 0, & x < d_{jt}, \end{cases} \quad (14)$$

with d_{jt} from Equation (9). The expected value of committing c , when a payment P has been received, is

$$V_{jt}(c, P) = \sum_{\mathbf{c}_{-j}} \Pr_{jt}(\mathbf{c}_{-j}) v_{jt}(b_{jt} + P + u_j(c, \mathbf{c}_{-j})), \quad (15)$$

and the automaton's voluntary commitment is

$$c_{jt}^* = \min \arg \max_{c \in \{0,1,2,3\}} V_{jt}(c, 0). \quad (16)$$

Ties are broken towards staying home.

The valuation is linear in goods above the survival threshold, so the automata are risk-neutral except for their wish to stay alive. The choice is made turn by turn: the automaton maximizes the expected value of the current turn and does not plan how its own commitment will change what the others expect of it. In autarky this costs it nothing (Section 5.6).

5.3 Consumption

After production every automaton follows the same buffer rule with a target reserve B :

$$z_{jt} = \min\{A_{jt}, \max\{d_{jt}, A_{jt} - B\}\}. \quad (17)$$

It consumes what it needs to survive the turn, keeps up to B goods in reserve, and consumes everything above that. With $B = 1$, as in both levels, an automaton in autarky consumes three per turn and holds one good (in Level 2 it first eats the second good it started with). An automaton that has had a good turn at the table eats most of the gain at once, which lowers its requirement in the following turn or turns.

5.4 Parameters

Automaton	Trade	Initial forecast	Learning rate η_j	Level
Sable	Mystic	(1, 0, 0, 0)	0.27	1, 2
Iona	Scientist	(1, 0, 0, 0)	0.22	1, 2
Marlow	Farmer	(1, 0, 0, 0)	0.17	1, 2
Ansel	Craftsperson	(1, 0, 0, 0)	0.12	1, 2
Mira	Healer	(1, 0, 0, 0)	0.27	2
Rowan	Organizer	(1, 0, 0, 0)	0.22	2

Table 5: The automata. They start with the same expectations and differ only in how fast they revise them.

5.5 What moves an automaton

At the start. With everybody forecast to stay home, any commitment is expected to be wasted, so every automaton's voluntary commitment is zero. This is the autarky equilibrium, held in place by beliefs.

	Contributors observed, m	Sable	Mira	Iona	Rowan	Marlow	Ansel
		.27	.27	.22	.22	.17	.12
$n = 5$	1	never		never		never	never
	2	4		4		6	8
	3	2		3		3	5
	4	2		2		3	3
$n = 7$	2	never	never	never	never	never	never
	3	4 (2)	4 (2)	5 (2)	5 (2)	7 (2)	10 (2)
	4	3 (2)	3 (2)	4	4	5 (2)	7 (2)
	5	2	2	3	3	4	5 (2)
	6	2	2	2	2	3	4 (2)

Table 6: Turns of observation after which an automaton that has stayed home first contributes voluntarily, when m others (the instigator among them) commit everything every turn. It then commits all three units, except where “(2)” marks a first commitment of two. “Never”: not within forty turns.

After observing contributions. Consider an automaton that has stayed home so far and has watched m other players commit everything, turn after turn, while the rest stayed home. Table 6 gives the number of such turns after which its voluntary commitment first becomes positive.

Three things stand out. First, the table reproduces the critical mass of Section 3.5: one committed partner never suffices with five players, nor two with seven, however long they persist, because the resulting table cannot beat home production. Second, the automata do not wait for certainty. With two committed partners in the five-player game, Sable joins when it believes each of them will show up again with probability $1 - 0.73^4 \approx 0.72$. Third, the order in which automata come round follows their learning rates, and the gap is large: Ansel needs twice as many observations as Sable.

The automata are more cautious than the break-even belief q^* of Section 3.5 suggests, because of survival. An automaton in autarky holds one good and needs three this turn. If it commits everything and the table fails, it has one good and dies. Expression (14) values that outcome at zero, which weighs against commitment until failure is sufficiently unlikely. In the seven-player game the automata often hedge and commit two units first, keeping one at home.

When contributors stop. Everything above runs in reverse at the same speed. If the partners an automaton has come to rely on stay home, its forecast of them decays by the factor $(1 - \eta_j)$ per turn, and the automaton withdraws once the expected value of committing falls below that of staying home. Its withdrawal is in turn observed by the others. Cooperation that rests on a few players can therefore unravel, and it unravels fastest among those who were quickest to join.

5.6 In what sense the automata are rational

The game asks whether individual rationality guarantees collective optimality. It therefore matters that the automata are rational, and that the trap they are in does not come from any limit on their reasoning.

They do what is best for themselves. Each automaton chooses the commitment that maximizes its own expected payoff, given what it expects the others to do (Equation 16). This is what individual rationality means in game theory. Nothing in an automaton’s choice is habit, imitation or error.

Their expectations are correct. A rational player still has to expect something of the others, and rationality does not say what. Both equilibria of Section 3 are consistent with every player

being rational and every player knowing that the others are. The automata take their expectations from experience. At the start everybody expects everybody to stay home, everybody stays home, and everybody was right. This is a Nash equilibrium in the full sense of the term: rational players whose expectations of each other are correct. Yet each of them would produce three and a half times as much in the other equilibrium.

More foresight would not help. The automata decide turn by turn. One may ask whether an automaton that planned ahead, and reckoned with the effect of its own contributions on what the others come to expect, could lead the society out of autarky. It could not. Whatever one player does, it is one contributor, and one contributor never brings a second to the table: staying home is the best reply even to a partner who is certain to commit everything (Table 4), so the others would not follow however long it persisted (the rows marked “never” in Table 6). Nor could it persist. Energy sent to an empty table is lost, and autarky leaves no surplus to lose: over its whole life a player can waste as many units of energy as it has goods in its starting reserve, one in Level 1 and two in Level 2, before it fails the survival check. In autarky, staying home is therefore the best an automaton can do however far ahead it looks. Away from autarky the horizon can matter: once others contribute, a player looking further ahead might join somewhat earlier or later than the automata do. That affects how fast a society moves, not whether rational players can be stuck.

What the instigator has is an instrument, not better judgment. Autarky cannot be left through one player’s contribution. It takes two contributors in Level 1, and three in Level 2, before another player finds it worthwhile to join. The instigator can bring this about because he or she can add a bought contribution to his or her own (Section 6), at a loss and at a risk to survival that no player would accept for the sake of the current turn.

Learning is the way out, not the cause. That the automata revise their expectations in the light of what they see is what makes leaving autarky possible at all. Players who held to their expectations whatever they observed would be no less rational, and could never be moved.

6 Pledges

The instigator has one way to influence the automata directly: paying them.

6.1 Timing and information

At the start of a turn each automaton fixes its voluntary commitment c_{jt}^* from the history so far, together with the prices at which it would add one, two or three further units of energy this turn (as far as its three units allow). The instigator sees these commitments and prices, then chooses his or her own commitment and which offers to take up. A purchase of k additional units from automaton j at price $P_{jt}(k)$ is called a *pledge*.

- Pledges are paid in goods from the instigator’s reserve, before production. The total must not exceed the reserve: $\sum_j P_{jt}(k_j) \leq b_{At}$. The turn’s own production cannot be used.
- The paid automaton commits $c_{jt}^* + k$. This is binding: there is no default on either side.
- A pledge lasts one turn. Nothing carries over.
- Prices are fixed for the turn. They do not react to the instigator’s own commitment or to which other offers are taken up.
- A pledge is private. The other automata see the commitment it produces, but not that it was paid for. In their forecasts a bought contribution counts exactly as a voluntary one.

The instigator thus moves last and with full knowledge of what the automata will do voluntarily. This is an advantage the automata do not have, and it is what makes the instigator an instigator

rather than one more player guessing.

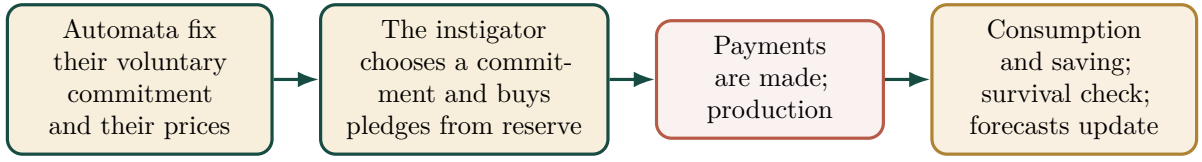


Figure 6: One turn.

6.2 The price

An automaton asks for the smallest payment that leaves it no worse off than its voluntary choice and certain to survive the turn. Payments are multiples of 0.05 goods. For $k \in \{1, \dots, 3 - c_{jt}^*\}$,

$$P_{jt}(k) = \min\{P \in \{0, 0.05, 0.10, \dots\} : \begin{aligned} &V_{jt}(c_{jt}^* + k, P) \geq V_{jt}(c_{jt}^*, 0) \\ &\text{and } b_{jt} + (3 - c_{jt}^* - k) + P \geq d_{jt} \end{aligned}\} \quad (18)$$

The first condition is about expected value: the payment must make up for what the automaton expects to lose by moving energy from home to the table. The second is about the worst case: reserve, remaining home production and payment must cover this turn's survival requirement *even if the table yields nothing*. A large enough payment satisfies both, so a living automaton always names a price.

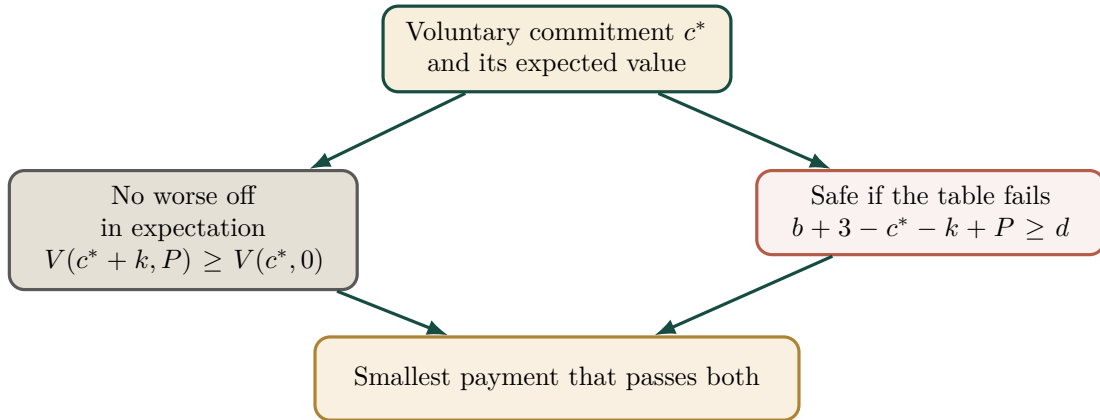


Figure 7: The two conditions on a pledge's price.

The price has the following properties.

- *It is a compensation, not a bargain.* The automaton does not ask what the pledge is worth to the instigator and adds no markup. It asks to be made whole.
- *It falls as expectations rise.* The more the automaton expects the others to contribute, the more it expects to earn at the table itself, and the less compensation it needs. As its expectations rise the price falls towards zero, and at the point where it would reach zero the automaton contributes unasked.
- *At the start it is one good per unit of energy.* With everybody forecast to stay home, each unit moved to the table is expected to be lost, and the automaton wants the home production it gives up: $P(k) = k$.
- *It is a price for a package.* Two units need not cost twice as much as one. The second unit changes the balance of the table, and it changes how much home production is left to fall back on.

- *Survival sets a floor.* From the turn in which the survival check first binds, an automaton in autarky holds one good and needs three, so its price for k units cannot fall below $k - 1$ however optimistic it is: the last unit of home production it gives up has to be replaced.
- *There is no ceiling.* The automaton's voluntary choice may carry a risk of death that it has accepted. A pledge must remove that risk for the turn, and the price of doing so can exceed the home production given up. For example, an automaton with a reserve of 1 that voluntarily commits two units and needs to consume 2.37 this turn keeps nothing at home if it sells its last unit. It asks 1.40 for that unit: the 1.37 it needs beyond its reserve, rounded up to the price grid.

6.3 Prices over time

Table 7 follows the fastest and the slowest learner in the five-player game while each watches two other players commit everything every turn.

Turns	Sable ($\eta = 0.27$)			Ansel ($\eta = 0.12$)		
	+1	+2	+3	+1	+2	+3
observed						
0	1.00	2.00	3.00	1.00	2.00	3.00
1	0.70	1.15	2.00	0.85	1.65	2.40
2	0.45	1.00	2.00	0.75	1.30	2.00
3	0.25	1.00	2.00	0.60	1.00	2.00
4	commits 3 unasked			0.50	1.00	2.00
5				0.40	1.00	2.00
6				0.35	1.00	2.00
7				0.25	1.00	2.00
8				commits 3 unasked		

Table 7: Prices for one, two and three additional units of energy, $n = 5$, for an automaton that has stayed home and watched two others commit everything.

The price of the first unit falls steadily. The prices of two and of three units fall until they meet the survival floor of $k - 1$ and stay there. Then, in a single step, the automaton commits everything voluntarily: once it believes in its partners enough to accept the risk, it wants to be at the table with all it has.

If only one other player commits, the price of the first unit also falls, to about 0.4, but the automaton never joins: a table of two cannot beat home production.

6.4 What a pledge costs

At the start of Level 1 the instigator holds one good, and every automaton asks one good per unit of energy. The reserve buys a single unit from a single automaton. If the instigator commits everything alongside it, the profile is the third row of Table 2: the table pays 0.634 per unit, and the instigator produces 1.90 instead of the 3 that staying home would have yielded. The first step thus costs the whole reserve and a third of a turn's production, and it leaves the instigator short of the survival requirement: with at most 1.90 consumed, at least 4.10 is needed in the second turn, more than home production can supply. The seller, by contrast, is well off: it has its payment and most of its home production.

This is the big push in miniature. The first mover bears a loss that only pays off if others follow, and has to stake survival on it.

7 The objective

7.1 What has to be achieved

The level is won at the end of a turn t in which

1. all n players are alive,
2. the instigator has committed all three units of energy, and
3. every automaton's voluntary commitment for the next turn, computed from its updated forecasts and its reserve after the turn, is three: $c_{j,t+1}^* = 3$ for all j .

In words: the society has reached the prosperous equilibrium, and it would stay there without any payment. There is no limit on the number of turns.

The condition is about willingness, not about one turn's outcome. A turn in which everybody is *paid* to commit everything produces 10.5 for everyone, but it is not a win unless the automata would do the same again unpaid.

7.2 The forces at work

The preceding sections imply the following.

Expectations are the state variable. The automata's choices depend on their forecasts, and their forecasts depend only on observed commitments. To move an automaton, the instigator has to change what that automaton *sees the others do*.

A pledge works on the bystanders. Paying an automaton changes its commitment this turn, and the commitment is seen by all the others, who cannot tell that it was bought. The paid automaton itself learns nothing from being paid; its own forecasts change only through what it observes of others. So a pledge to one automaton is an argument addressed to all the rest.

Critical mass comes first. A table of fewer than three contributors of five, or four of seven, pays less than home, and those at it lose goods every turn. An automaton joins voluntarily only when it expects to be at least the third, or the fourth: it must have watched two others, or three, contribute regularly (Table 6). Until then every contribution other than the instigator's own has to be bought. Above critical mass the table produces a surplus that can replenish the reserve.

Subsidy has to become unnecessary. Autarky generates no surplus, so the starting reserve cannot pay for many turns of pledges. Cooperation has to start financing itself, and then sustaining itself, before the reserve and the instigator's survival window run out.

Withdrawing support is observed too. When a paid contribution stops, the others see an absence, and their forecasts decay. Prices that had fallen rise again.

The automata are not alike. They respond at different speeds (Table 5). Who is approached first, and who is left to come round by watching, matters.

Mistakes can be permanent. An automaton that commits voluntarily and is let down may fail its survival check, and its trade is gone. A pledge protects the automaton it is paid to for the turn in question; it does not protect the others.

7.3 The larger society

Level 2 changes three things. There are seven trades, so the critical mass is four instead of three, a coalition of any given size is worth less (Table 1), and the break-even belief is higher. In compensation, every player starts with two goods instead of one, and survival is judged over three turns ($M = 9$, $L = 3$) instead of two, so a bad turn can be made up over two further turns rather than one. The required average of three per turn is unchanged.

8 A coordinating institution

The game also puts a different solution to the player, as a proposal that the players could vote on. Instead of one player shifting expectations at private cost, the society founds an institution with a single rule:

Every unit of energy at the table earns at least $g = 1.01$. If the table yields less, the difference is paid to the contributors and financed in the same turn by a proportional tax on what each player has produced.

Formally, with $\tau(\mathbf{c}) = (g - R(\mathbf{c}))^+ T(\mathbf{c}) / \sum_i u_i(\mathbf{c})$ the tax rate that balances the budget, player i receives

$$\tilde{u}_i(\mathbf{c}) = (3 - c_i) + c_i \max\{R(\mathbf{c}), g\} - \tau(\mathbf{c}) u_i(\mathbf{c}). \quad (19)$$

Exhaustive enumeration of this game, for $n = 5$ and $n = 7$ with all players alive, shows:

1. *Staying home is never a best reply*, whatever the others do. In particular universal autarky is no longer an equilibrium: a player who commits everything while all others stay home receives $3g = 3.03$ and pays no tax, because the tax falls on production and an unsuccessful contributor has produced nothing.
2. *Universal full commitment is the only pure-strategy Nash equilibrium*. With five players the game is even dominance-solvable: staying home is strictly dominated, then committing one unit, then two.
3. *In that equilibrium the guarantee costs nothing*. The table pays 3.5, far above g ; nobody is topped up and nobody is taxed.

The institution removes the bad equilibrium by insuring against it, and because the insurance is believed, it is never called upon. This is the logic of deposit insurance against bank runs [10] and of the dominant assurance contract [30]; the name the game gives the institution, the Commonwealth, follows Hobbes [15], for whom a commonwealth is what people institute by agreement to escape a condition in which each acting alone leaves all worse off.

Two qualifications are worth stating.

Who pays matters. The result depends on the tax falling on production. If the same guarantee were financed by a tax on incomes including the top-up, or by an equal levy per head, a lone contributor would bear a share of his own subsidy large enough to make entering unprofitable (about 2.4 instead of 3 in the five-player game), and universal autarky would remain an equilibrium. The guarantee has to be paid for by those who stay home.

The institution cannot outlive the economy. The guarantee is worth founding only while full cooperation among the living beats autarky. Once deaths have reduced the society to two of five trades, or three of seven, the full table pays less than g (Table 1); the top-ups would have to be paid every turn out of the same players' production, everybody would be worse off than at home, and nobody would vote for it.

The margin above the home return can be arbitrarily small, but it cannot be zero. With $g = 1$ a lone contributor would only break even, and universal autarky would survive as an equilibrium.

9 Scope and limitations

What is proved. The equilibrium results of Section 3 concern pure strategies in the one-turn game without payments, and those of Section 8 the one-turn game with the guarantee. Both are established by complete enumeration. Mixed equilibria are not characterized.

How the dynamic game is treated. In game theory a dynamic game counts as solved when an equilibrium of the whole game has been found: a complete plan for every player, covering every situation that can arise, such that each plan is a best reply to the others' plans in each of those situations [27, 20]. A solution of this kind takes for granted that the players anticipate each other's plans correctly. That is exactly the knowledge the players of this game do not have, and what happens without it is what the game is about. An equilibrium analysis would also have little to add. Staying home forever is an equilibrium of the repeated game, pledges included: against players who go on expecting autarky, and price their energy accordingly, the instigator's reserve never buys a table that pays. Committing everything forever is another. Solving the dynamic game in this sense would therefore return the two states of Section 3 and say nothing about how a society gets from one to the other. The paper takes the other established route, the theory of learning in games [12]: players who do not know what the others will do form their expectations from what they observe, and act on them as well as they can. This is not a shortcut. That rational players who are uncertain about one another cannot in general both choose optimally and predict each other correctly is a known result [23, 11].

What is left open. First, the best way to play. Because the automata follow known rules, the instigator faces a well-defined problem of planning over time: which sequence of commitments, pledges and consumption wins in the fewest turns, or with the highest score. The paper does not answer it; that is the task the game sets the player. Second, the automata away from autarky. Section 5.6 shows that choosing turn by turn is the best an automaton can do while everybody else stays home. Whether a player planning further ahead could do better once others contribute is not examined.

Simplifications in the automata. They choose turn by turn, which costs them nothing in autarky but can matter elsewhere (Section 5.6). They treat the other players' commitments as independent, although real commitments rise and fall together. They read a contribution as evidence about the contributor and do not ask why it was made. And they all save by the same simple rule. A faster learner is quicker to trust and quicker to give up; there is no separate parameter for either.

What the parameters mean. The return of 3.5, the exponent of 1.5, the survival requirement and the learning rates are design choices that make the coordination problem visible and playable. They are not estimates of anything. What carries over from the game is the structure: a division of labour that pays only if enough others take part, an economy that can rest in either state, and the difference between being better off in the good state and being able to get there.

A Parameters

	Level 1	Level 2
Players n	5	7
Energy per player and turn	3	3
Home return per unit	1	1
Table return per unit	$3.5 (H/5)^{1.5}$	$3.5 (H/7)^{1.5}$
Smallest group that beats autarky	3	4
Starting reserve b_0	1	2
Survival: minimum M over L turns	6 over 2	9 over 3
Automata's target reserve B	1	1
Automata's initial forecast π_0	(1, 0, 0, 0)	(1, 0, 0, 0)
Learning rates η_j	Sable .27, Iona .22, Marlow .17, Ansel .12	as Level 1; Mira .27, Rowan .22
Price grid for pledges	0.05	0.05
Guarantee g (Section 8)	1.01	1.01

Table 8: Parameter values.

B The sequence of a turn

1. Every living player receives three units of energy. Reserves are carried over from the previous turn.
2. Each automaton computes its voluntary commitment (Equation 16) and its prices for additional energy (Equation 18) from the history so far.
3. The instigator, knowing these, chooses a commitment and any pledges the reserve can pay for.
4. Payments are transferred. Each paid automaton's commitment is raised by the units bought.
5. Production takes place (Equations 1–3).
6. If the instigator's available goods fall short of the survival requirement, the run ends. Otherwise the instigator chooses consumption between the requirement and the goods available; the automata consume by the buffer rule (Equation 17). What is not consumed is carried over.
7. Survival is checked for every player (Equation 8). Automata that fail are removed.
8. Every automaton updates its forecasts from the commitments observed this turn (Equation 12).
9. The winning condition is evaluated. If it is not met, the next turn begins. Pledges do not carry over.

References

- [1] Gary S. Becker and Kevin M. Murphy. The division of labor, coordination costs, and knowledge. *Quarterly Journal of Economics*, 107(4):1137–1160, 1992.
- [2] George W. Brown. Iterative solution of games by fictitious play. In T. C. Koopmans (ed.), *Activity Analysis of Production and Allocation*, pages 374–376. Wiley, 1951.

- [3] John Bryant. A simple rational expectations Keynes-type model. *Quarterly Journal of Economics*, 98(3):525–528, 1983.
- [4] Hans Carlsson and Eric van Damme. Global games and equilibrium selection. *Econometrica*, 61(5):989–1018, 1993.
- [5] Yin-Wong Cheung and Daniel Friedman. Individual learning in normal form games: some laboratory results. *Games and Economic Behavior*, 19(1):46–76, 1997.
- [6] Russell W. Cooper. *Coordination Games: Complementarities and Macroeconomics*. Cambridge University Press, 1999.
- [7] Russell Cooper and Andrew John. Coordinating coordination failures in Keynesian models. *Quarterly Journal of Economics*, 103(3):441–463, 1988.
- [8] Vincent P. Crawford. Adaptive dynamics in coordination games. *Econometrica*, 63(1):103–143, 1995.
- [9] Peter A. Diamond. Aggregate demand management in search equilibrium. *Journal of Political Economy*, 90(5):881–894, 1982.
- [10] Douglas W. Diamond and Philip H. Dybvig. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy*, 91(3):401–419, 1983.
- [11] Dean P. Foster and H. Peyton Young. On the impossibility of predicting the behavior of rational agents. *Proceedings of the National Academy of Sciences*, 98(22):12848–12853, 2001.
- [12] Drew Fudenberg and David K. Levine. *The Theory of Learning in Games*. MIT Press, 1998.
- [13] John C. Harsanyi and Reinhard Selten. *A General Theory of Equilibrium Selection in Games*. MIT Press, 1988.
- [14] Benjamin E. Hermalin. Toward an economic theory of leadership: leading by example. *American Economic Review*, 88(5):1188–1206, 1998.
- [15] Thomas Hobbes. *Leviathan*. London, 1651.
- [16] Michihiro Kandori, George J. Mailath, and Rafael Rob. Learning, mutation, and long run equilibria in games. *Econometrica*, 61(1):29–56, 1993.
- [17] Michael Kremer. The O-ring theory of economic development. *Quarterly Journal of Economics*, 108(3):551–575, 1993.
- [18] Paul Krugman. History versus expectations. *Quarterly Journal of Economics*, 106(2):651–667, 1991.
- [19] Ik Soo Lim, Peter Wittek, and John Parkinson. On the origin of risk sensitivity: the energy budget rule revisited. *Animal Behaviour*, 110:69–77, 2015.
- [20] Eric Maskin and Jean Tirole. Markov perfect equilibrium: I. Observable actions. *Journal of Economic Theory*, 100(2):191–219, 2001.
- [21] Kiminori Matsuyama. Increasing returns, industrialization, and indeterminacy of equilibrium. *Quarterly Journal of Economics*, 106(2):617–650, 1991.
- [22] Kevin M. Murphy, Andrei Shleifer, and Robert W. Vishny. Industrialization and the big push. *Journal of Political Economy*, 97(5):1003–1026, 1989.
- [23] John H. Nachbar. Prediction, optimization, and learning in repeated games. *Econometrica*, 65(2):275–309, 1997.

- [24] Paul N. Rosenstein-Rodan. Problems of industrialisation of Eastern and South-Eastern Europe. *Economic Journal*, 53(210/211):202–211, 1943.
- [25] Jean-Jacques Rousseau. *Discours sur l'origine et les fondements de l'inégalité parmi les hommes*. Amsterdam, 1755.
- [26] A. D. Roy. Safety first and the holding of assets. *Econometrica*, 20(3):431–449, 1952.
- [27] Reinhard Selten. Reexamination of the perfectness concept for equilibrium points in extensive games. *International Journal of Game Theory*, 4(1):25–55, 1975.
- [28] Brian Skyrms. *The Stag Hunt and the Evolution of Social Structure*. Cambridge University Press, 2004.
- [29] Adam Smith. *An Inquiry into the Nature and Causes of the Wealth of Nations*. London, 1776.
- [30] Alexander Tabarrok. The private provision of public goods via dominant assurance contracts. *Public Choice*, 96(3–4):345–362, 1998.
- [31] John B. Van Huyck, Raymond C. Battalio, and Richard O. Beil. Tacit coordination games, strategic uncertainty, and coordination failure. *American Economic Review*, 80(1):234–248, 1990.
- [32] H. Peyton Young. The evolution of conventions. *Econometrica*, 61(1):57–84, 1993.
- [33] Allyn A. Young. Increasing returns and economic progress. *Economic Journal*, 38(152):527–542, 1928.